

CS771 Assignment 1

Breaking the Hamming PUF using Linear Models

Annadevara Yashvanth Chary

1 Question 1: Linear Representation of the Hamming PUF

1.1 Problem Setup

The Hamming PUF stores a secret word

$$s = (s_0, s_1, \dots, s_{31}) \in \{0, 1\}^{32}.$$

Given a challenge

$$c = (c_0, c_1, \dots, c_{31}) \in \{0, 1\}^{32},$$

the PUF computes

$$m = c \oplus s.$$

The partial Hamming weights are

$$h_e = \sum_{i \in E} m_i,$$

$$h_o = \sum_{j \in O} m_j,$$

where

$$E = \{0, 2, 4, \dots, 30\}$$

and

$$O = \{1, 3, 5, \dots, 31\}.$$

The response is

$$r(c) = \begin{cases} 1 & \text{if } h_e h_o \geq t \\ 0 & \text{otherwise.} \end{cases}$$

Our goal is to represent this decision rule using a linear classifier

$$r(c) = \frac{1 + \text{sign}(w^T \phi(c) + b)}{2}.$$

1.2 Rewriting XOR

For binary bits,

$$m_i = c_i \oplus s_i.$$

Since s_i is fixed,

$$m_i = \begin{cases} c_i, & s_i = 0 \\ 1 - c_i, & s_i = 1. \end{cases}$$

This can be written as

$$m_i = (1 - 2s_i)c_i + s_i.$$

Define

$$a_i = 1 - 2s_i,$$

$$b_i = s_i.$$

Then

$$m_i = a_i c_i + b_i.$$

Notice that a_i and b_i depend only on the secret word and are therefore constants for a fixed PUF.

1.3 Expanding $h_e h_o$

We have

$$h_e h_o = \left(\sum_{i \in E} m_i \right) \left(\sum_{j \in O} m_j \right).$$

Expanding,

$$h_e h_o = \sum_{i \in E} \sum_{j \in O} m_i m_j.$$

Substituting $m_i = a_i c_i + b_i$,

$$h_e h_o = \sum_{i \in E} \sum_{j \in O} (a_i c_i + b_i)(a_j c_j + b_j).$$

Expanding further,

$$h_e h_o = \sum_{i \in E} \sum_{j \in O} (a_i a_j c_i c_j + a_i b_j c_i + a_j b_i c_j + b_i b_j).$$

The above expression contains:

- Quadratic terms $c_i c_j$

- Linear terms c_i
- Constant terms

Hence

$$h_e h_o = \sum_{i \in E, j \in O} A_{ij} c_i c_j + \sum_{k=0}^{31} B_k c_k + C,$$

where A_{ij} , B_k and C are constants depending only on the secret word s .

1.4 Feature Map

The final expression contains only linear and quadratic terms.

Therefore define

$$\phi(c) = (c_0, \dots, c_{31}, \{c_i c_j : i \in E, j \in O\}).$$

The first 32 coordinates are the original challenge bits.

The remaining coordinates are all pairwise products between even and odd challenge bits.

1.5 Linear Model

Choose the weight vector w such that

$$w_k = B_k$$

for linear features and

$$w_{ij} = A_{ij}$$

for quadratic features.

Then

$$w^T \phi(c) = \sum_{i \in E, j \in O} A_{ij} c_i c_j + \sum_{k=0}^{31} B_k c_k.$$

Choose

$$b = C - t.$$

Therefore

$$w^T \phi(c) + b = h_e h_o - t.$$

Hence

$$w^T \phi(c) + b \geq 0$$

if and only if

$$h_e h_o \geq t.$$

Thus

$$r(c) = \frac{1 + \text{sign}(w^T \phi(c) + b)}{2}.$$

1.6 Conclusion

The Hamming PUF can be represented exactly by a linear classifier using the feature map

$$\phi(c) = (c_0, \dots, c_{31}, \{c_i c_j : i \in E, j \in O\}).$$

2 Question 2: Dimensionality of the Feature Map

The feature map contains:

1. 32 linear features corresponding to the challenge bits.
2. Quadratic features $c_i c_j$ where i is even and j is odd.

There are 16 even positions and 16 odd positions.
Therefore the number of quadratic features is

$$16 \times 16 = 256.$$

Hence the total dimensionality is

$$D = 32 + 256 = 288.$$

The bias term is represented separately and therefore no additional constant feature is required.

$$\boxed{D = 288}$$

3 Question 3: Implementation and Hyperparameter Selection

The feature map was implemented by augmenting the original challenge vector with all pairwise products between even-indexed and odd-indexed challenge bits.

The mapped feature space has dimensionality

$$288.$$

Several LinearSVC configurations were evaluated on the public dataset.

The final selected hyperparameters were:

```
C = 2  
loss = hinge  
dual = True  
tol = 0.01  
max_iter = 20000
```

This configuration achieved:

- Training Accuracy = 99.91%
- Test Accuracy = 99.80%

which was the highest observed public test accuracy.

4 Question 4: Experimental Evaluation on the Public Dataset

The proposed 288-dimensional feature map was used in all experiments.

4.1 Effect of C in LinearSVC

C	Test Accuracy (%)	Training Time (s)
0.01	97.40	0.0835
0.1	98.72	0.0925
1	99.68	0.0945
10	99.60	0.1084
100	99.60	0.1263

Increasing C improved accuracy up to $C = 1$, after which performance saturated.

4.2 Effect of Loss Function in LinearSVC

Loss Function	Test Accuracy (%)	Training Time (s)
hinge	99.72	0.1673
squared_hinge	99.68	0.1033

The hinge loss achieved slightly higher accuracy while squared hinge trained faster.

4.3 Effect of Tolerance in LinearSVC

tol	Test Accuracy (%)	Training Time (s)
10^{-2}	99.36	0.0792
10^{-4}	99.68	0.1025
10^{-6}	99.68	0.1108

Reducing tolerance improved accuracy up to 10^{-4} , beyond which no improvement was observed.

4.4 Effect of C in Logistic Regression

C	Test Accuracy (%)	Training Time (s)
0.01	96.32	0.0643
0.1	97.72	0.0902
1	98.68	0.1621
10	99.32	0.1233
100	98.44	0.0910

The best Logistic Regression performance was achieved at $C = 10$.

4.5 Effect of Penalty in Logistic Regression

Penalty	Test Accuracy (%)	Training Time (s)
L1	98.64	1.5059
L2	98.36	0.1807

L1 regularization achieved slightly better accuracy but required significantly more training time.

4.6 Comparison of LinearSVC and Logistic Regression

Model	Best Test Accuracy (%)
LinearSVC	99.80
Logistic Regression	99.32

LinearSVC consistently outperformed Logistic Regression on the public dataset.

5 Question 5: Effect of Training Set Size

The selected LinearSVC model used:

- $C = 2$
- loss = hinge
- dual = True
- tol = 0.01
- max_iter = 20000

5.1 Results

Training Set Size	Test Accuracy (%)
100	80.04
250	88.00
500	89.60
750	91.96
1000	93.28
1500	93.88
2000	95.40
3000	97.72
4000	98.20
5000	99.36
7500	99.80

5.2 Discussion

The accuracy increases steadily as the training size increases. Significant gains occur between 1000 and 3000 CRPs, while performance begins to saturate beyond 4000 CRPs.

5.3 Minimum Number of CRPs Required

Target Accuracy	Required CRPs
95%	2000
97%	3000
99%	5000

5.4 Conclusion

The Hamming PUF is highly vulnerable to machine-learning attacks. With only 5000 CRPs, the model exceeds 99% test accuracy, while the full training set of 7500 CRPs achieves 99.80% accuracy.